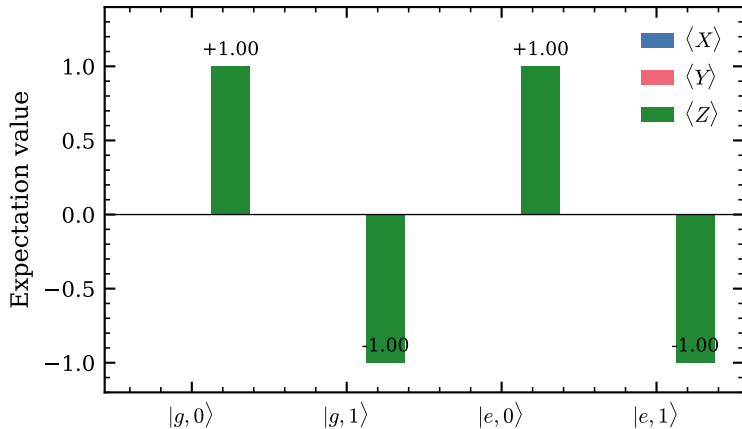


Appendix A2 — Target output state observables for each logical input

$$U_{\text{target}} = \text{SWAP} \cdot \text{CZ} \cdot (H \otimes I), \text{ basis } \{|g, 0\rangle, |g, 1\rangle, |e, 0\rangle, |e, 1\rangle\}$$

Transmon Bloch-vector components
(output of U_{target} , exact)



Cavity Fock populations
(output of U_{target} , exact)

